



Modeling of the dynamic adsorption of an anionic dye through ion-exchange membrane adsorber

Jordi Labanda^{a,*}, Josep Sabaté^b, Joan Llorens^a

^a Department of Chemical Engineering, University of Barcelona, Martí i Franquès 1, 08028 Barcelona, Spain

^b Departament d'Enginyeria Agroalimentària i Biotecnologia, Universitat Politècnica de Catalunya, Edifici ESAB, Avinyuda del Canal Olímpic s/n, 08860 Castelldefels, Spain

ARTICLE INFO

Article history:

Received 22 October 2008

Received in revised form 30 March 2009

Accepted 23 May 2009

Available online 30 May 2009

Keywords:

Membrane chromatography
Ion-exchange membrane adsorber
Dynamic adsorption
Breakthrough curve
Anionic dye

ABSTRACT

The present study examines the dynamic adsorption through ion-exchange membrane adsorbers. The model used in the study includes convection, axial dispersion with simultaneous adsorption and desorption of the solute in the membrane. Adsorption and desorption processes give the Langmuir isotherm for the equilibrium. The mathematical model makes use of dimensionless parameters in terms of characteristic times for the different mechanisms that take place during the process (convection, dispersion, adsorption and desorption characteristic times). The model has five independent dimensionless parameters. Three of these parameters are related to the equilibrium isotherm and the other two are related to the dynamic process. Equilibrium and dynamic experiments were carried out in order to fit their respective parameters. In order to examine the suitability of the model to describe real processes, the adsorption of an anionic dye (Orange-G) through the ion-exchange membrane adsorber was investigated as a function of dye and KCl concentration, obtaining strong correlation between fitted and experimental breakthrough curves. The results show the relative importance of axial dispersion, adsorption and desorption as a function of operational variables.

© 2009 Elsevier B.V. All rights reserved.

1. Introduction

Wastewater from textile dyeing and finishing factories is a significant source of environmental pollution. Textile wastewater is characterized by high chemical oxygen demand, low biodegradability, and high salt content and is the source of aesthetic pollution related to color. In addition, it has been stressed recently that color compounds such as dyes and pigments discharged from several industries are very harmful to aquatic life in rivers and lakes. Therefore, it is of great importance to eliminate them from the effluents not only for the removal of the color from water but also for environmental preservation. However, most of them are not oxidized by conventional, biological, and physical treatments because of their complex structure and larger molecular size [1–3].

Adsorption appears to have considerable potential for the removal of color from industrial effluents. Many researchers have followed up the possibility of an adsorption process to remove color compounds with ion-exchange membranes, and their advantages for controlling water pollution are less space, lower capital investment, simple design, easy operation of the equipment and efficient removal of organic waste constituents in comparison with other traditional treatment process [4].

Adsorption and ion-exchange processes are by far the most widely used techniques for high-resolution separation and analysis of dyes. These processes are traditionally carried out using packed beds, which have several major limitations. The pressure drop across a packed bed is generally high. Another major limitation with conventional adsorption and ion-exchange processes is the dependence on intra-particle diffusion for the transport of solute molecules to their binding sites within the pores of such media [5–7]. This increases the process time since transport of big molecules by diffusion is slow, and more when it is hindered. Consequently, the recovery liquid volume needed for elution also increases. Some of these factors and the fact that the transport phenomenon is complicated make scale-up of packed bed adsorption and ion-exchange processes difficult.

Recently, a new technology in membrane separation, the membrane chromatography, has proven its efficiency and time stability. Membrane chromatography has been studied as an alternative to conventional resin-based adsorption columns for the isolation and purification of molecules. Membrane chromatography combines the advantages of conventional chromatography columns in terms of separation power and capacity with membrane technology regarding mass transfer, high throughput and robustness. The main advantage is the absence of long diffusive paths, thus a process performed with membrane adsorber is definitely faster than with a traditional column configuration [8]. Hence, membrane chromatography operates in convective mode, which can significantly

* Corresponding author. Tel.: +34 934021312; fax: +34 934021291.
E-mail address: jlbanda@ub.edu (J. Labanda).

reduce diffusion and pressure drop limitations commonly encountered in column separation processes [9,10]. However, diffusive transport is not totally absent and convective flow of inappropriate type can be a serious disadvantage. Rational design of the membrane chromatography process and equipment is possible only when the transport phenomena involved are properly understood.

Membrane chromatography is performed by four types of membrane adsorbers: ion-exchange, affinity, hydrophobic interaction and reversed-phase. The uses of ion-exchange and affinity interactions are more widely reported in the literature [11]. Affinity membranes has attached onto the membrane surface different ligands that are able to immobilize specific molecules, such as proteins and biomolecules [12,13]. For example, dyes can function as an ion-exchange to bind proteins by electrostatic interactions [14–17]. Nevertheless, ion-exchange membranes retain directly charged molecules and their ion-exchange capacity is given by the introduction to the solid-phase of different charged groups such as sulfonic acid, sulfopropyl, diethylaminoethyl and quaternary ammonium. Ion-exchange membranes have been also used for the purification of biomolecules [18,19].

The aim of the present study is to analyze and model for the dynamic adsorption of dye anions with sulfonic groups in a commercial flat-sheet ion-exchange membrane in respond to flux, dye and KCl concentrations and pH changes. In literatures, very few complete models have been proposed so far to describe the adsorption process in membrane chromatography, and the majority is obtained by adapting classical chromatography models. The optimization and scale-up of membrane chromatography in the recovery, separation and purification of chemical components is of major industrial importance [20]. The development of mathematical models to describe this membrane process and the use of these models to predict membrane performance are an engineering approach that can help to attain the engineering tasks successfully [21,22].

2. Theory

The movement of the solute in a membrane adsorber of surface A , thickness L , and porosity ε , is described by the differential mass balance of solute in a slice of membrane with the following components: (1) the time derivation of solute concentration in the liquid-phase, (2) the axial convective flux, (3) the axial solute dispersion, which depends of the dispersion coefficient D_L , and (4) the solute adsorption/desorption onto the porous surface of the solid-phase as a function of time. The complete mass balance is expressed by the following equation with the corresponding initial and boundary conditions [23]:

$$\varepsilon \cdot \frac{\partial c}{\partial t} + \varepsilon \cdot v \cdot \frac{\partial c}{\partial z} = \varepsilon \cdot D_L \cdot \frac{\partial^2 c}{\partial z^2} - (1 - \varepsilon) \cdot \frac{\partial c_s}{\partial t} \quad (1)$$

$$c = 0 \quad \text{at } z \geq 0, t = 0. \quad (2)$$

$$c_s = 0 \quad \text{at } z \geq 0, t = 0 \quad (3)$$

$$\varepsilon \cdot v \cdot c - \varepsilon \cdot D_L \frac{\partial c}{\partial z} = \varepsilon \cdot v \cdot c_0 \quad \text{at } z = 0, t > 0 \quad (4)$$

$$\frac{\partial c}{\partial z} = 0 \quad \text{at } z = L, t > 0 \quad (5)$$

where c is the solute concentration in the liquid-phase, c_s is the adsorbed solute concentration in the solid-phase, v is the axial velocity of the liquid-phase and D_L is the axial dispersion coefficient for the solute. Hydrodynamic dispersion in flow through porous media may depend on molecular diffusion, D_m , axial velocity, v , and the morphology of the porous medium, which can strongly affect dispersion process [24]. Often, $D_L > D_m$ because of (a) the Taylor dispersion, as a competition between convective mixing and

molecular diffusion, (b) local turbulent diffusion, and (c) differences in local axial velocity into the porous media.

In this work, the adsorption/desorption equilibrium for the solute distribution between liquid-phase and solid-phase is described by the Langmuir isotherm. The Langmuir isotherm assumes energetic homogeneity of the adsorption sites without steric effects. The Langmuir isotherm involves a second order kinetic reversible process between adsorption and desorption:

$$\frac{\partial c_s}{\partial t} = k_a \cdot c \cdot (c_s^* - c_s) - k_d \cdot c_s \quad (6)$$

where c_s^* is the maximum possible concentration of adsorbed solute in the solid-phase and k_a and k_d are the adsorption and desorption rate constants, respectively. For a liquid-phase concentration c_{eq} , the equilibrium solute concentration in the solid-phase $c_{s,eq}$, can be deduced from Eq. (6) when $(\partial c/\partial t) = 0$ and the Langmuir isotherm appears:

$$c_{s,eq} = \frac{b \cdot c_s^* \cdot c_{eq}}{1 + b \cdot c_{eq}} \quad (7)$$

where b is the ratio between the adsorption and desorption rate constants, $b = k_a/k_d$.

The above model has been solved in terms of dimensionless parameters [23,25,26]. Now, we propose a redefinition of some dimensionless parameters in terms of four characteristic times related with important mechanisms in the process: the average retention time or characteristic time of the process, $t_{process}$, the axial dispersion time, t_{disp} , and the adsorption and desorption times, t_{ads} and t_{des} , defined as follows:

$$t_{process} = \frac{L}{v} \quad (8)$$

$$t_{disp} = \frac{L^2}{D_L} \quad (9)$$

$$t_{ads} = \frac{1}{k_a \cdot c_0} \quad (10)$$

$$t_{des} = \frac{1}{k_d} \quad (11)$$

The comparative importance of the dispersion, adsorption and desorption mechanisms in the process are directly related to their characteristics times in comparison with the convection time. If a mechanism has a characteristic time lower than the convective time, this mechanism has time to take place in the membrane and it is significant in the performance of the global process and vice versa. The radial diffusive time is assumed to be several orders of magnitude lower than the process time [26,27]. Therefore, radial diffusion eliminates the radial concentration gradients and only axial concentration gradients are considered in the model.

Relations between the characteristic times provide the following characteristic dimensionless parameters:

$$Pe = \frac{t_{disp}}{t_{process}} = \frac{v \cdot L}{D_L} \quad (12)$$

$$K_{ads} = \frac{t_{process}}{t_{ads}} = \frac{k_a \cdot c_0 \cdot L}{v} \quad (13)$$

$$r = \frac{t_{des}}{t_{ads}} = \frac{k_a}{k_d} \cdot c_0 \quad (14)$$

where Pe is the axial Peclet number, K_{ads} is a dimensionless adsorption constant and r is a relation between adsorption and desorption processes.

Eq. (1) can be rewritten in terms of these dimensionless parameters:

$$\frac{\partial C}{\partial \tau} + \frac{\partial C}{\partial \zeta} = \frac{1}{Pe} \cdot \frac{\partial^2 C}{\partial \zeta^2} - m \cdot \frac{\partial C_s}{\partial \tau} \quad (15)$$

$$m = \frac{1 - \varepsilon}{\varepsilon} \cdot \frac{C_{s,eq,0}}{C_0} \quad (16)$$

where C is the ratio between the punctual and the feed solute concentration in the liquid-phase, $C = c/c_0$, C_s is the ratio of concentrations in the solid-phase between the punctual and the equilibrium with c_0 , $C_s = c_s/c_{s,eq,0}$, ζ is a dimensionless coordinate along the membrane, $\xi = z/L$, and τ is a dimensionless time defined as the ratio between the current time and the convection time, $\tau = t/t_{process}$.

The kinetic and isotherm equations, Eqs. (6) and (7), can also be rewritten in dimensionless form:

$$\frac{\partial C_s}{\partial \tau} = K_{ads} \cdot \left(C + \frac{1}{r} \right) \cdot (C_{s,eq} - C_s) \quad (17)$$

$$C_{s,eq} = \frac{(r/p) \cdot C_{eq}}{1 + r \cdot C_{eq}} \quad (18)$$

where

$$C_{eq} = \frac{c_{eq}}{C_0} \quad (19)$$

$$p = \frac{C_{s,eq,0}}{C_s^*} \quad (20)$$

$$C_{s,eq,0} = \left(\frac{r}{1+r} \right) \cdot C_s^* \quad (21)$$

Now, the corresponding initial and boundary conditions are

$$C = 0 \quad \text{at } \zeta \geq 0, \tau = 0 \quad (22)$$

$$C_s = 0 \quad \text{at } \zeta \geq 0, \tau = 0 \quad (23)$$

$$C - \frac{1}{Pe} \frac{\partial C}{\partial \zeta} = C_0 \quad \text{at } \zeta = 0, \tau > 0 \quad (24)$$

$$\frac{\partial C}{\partial \zeta} = 0 \quad \text{at } \zeta = 1, \tau > 0 \quad (25)$$

The model is governed by Eqs. (15)–(25), which were implemented in the program “Mathematica” (Wolfram Research) in order to solve the differentials. As a result, the dynamic adsorption, represented by the breakthrough curve, of a solute through the membrane adsorber can be simulated and compared to experimental data.

The model depends on five independent dimensionless parameters: r , p , m , Pe and K_{ads} . Three of these parameters (r , p and m) are related to the equilibrium isotherm and the other two (Pe and K_{ads}) are related to the dispersion and the adsorption/desorption dynamic process. Equilibrium and dynamic experiments were carried out in order to fit the respective parameters.

The dimensionless parameters r , p and m can be calculated from the parameter b and C_s^* of Langmuir isotherm as follows:

$$r = b \cdot C_0 \quad (26)$$

$$p = \frac{r}{1+r} \quad (27)$$

$$m = \frac{1 - \varepsilon}{\varepsilon} \cdot \frac{b \cdot C_s^*}{1 + b \cdot C_0} \quad (28)$$

The rest of parameters (Pe and K_{ads}) can be determined by matching the numerical solution with the experimental breakthrough curves, where the outlet solute concentration is represented as a function of time.

The parameter m defines deloading capacity of the membrane as can be seen from Eq. (16). As m increases, the breakthrough curve is shifted gradually to higher dimensionless times. At the same time, as Pe increases axial dispersion decreases and the breakthrough curves become sharper. On the other side, K_{ads} is the ratio between the process time and adsorption characteristic time. A large K_{ads} corresponds to a fast adsorption process or a large time process. If

$K_{ads} \ll 1$ adsorption is tiny and breakthrough curve starts at lower τ values. If $K_{ads} \gg 1$ adsorption is complete and breakthrough curve starts at higher τ values.

The model is a general expression valid to describe the adsorption process in a porous media, as a membrane adsorber, where the radial diffusion can be neglected. Eqs. (1) and (15) can be simplified when the axial dispersion does not take place during the convective time ($Pe > 250$) [28]. In this case, the model presents analytical solution, which relates the solute concentration with the adsorption constant and time [29].

3. Experimental

The adsorption of the anionic dye Orange-G (Sigma–Aldrich, Germany) through an ion-exchange membrane adsorber Sartobind D (Sartorius, Germany) has been investigated to evaluate the model predictions. The membrane is formed by 3 flat-sheets of weak anion exchanger with diethylamine as a functional group. The features of this membrane are 25 mm of membrane diameter (4.91 cm² of membrane area), an average membrane thickness of about 0.8 mm and a porosity, ε , equal to 0.78 [30]. The anionic dye Orange-G (Color Index Number = 16,230) is an azo synthetic dye with the following features: MW = 452.37 g/mol and λ_{max} = 478 nm.

The anionic dye feed solutions were prepared by solubilizing the powder dye in a buffer solution at pH = 7 with three different KCl concentrations. The buffer solution was prepared by adding 305 mL of Na₂HPO₄ solution 0.2 M and 195 mL of NaH₂PO₄ solution 0.2 M into 1 L of deionized water. Then, KCl was added in order to have 0.1, 0.5 and 1.0 M KCl solutions.

Solutions were fed into the membrane cartridge by a syringe pump NE-1000 (New Era Pump Systems Inc., New York) set at a constant flow rate of 2.0, 1.0, 0.5 and 0.1 mL min^{−1}. The outlet dye concentrations were determined discontinuously every 5 mL using a UV–vis spectrophotometer (Secoman, France). All experiments were conducted at a constant temperature, 25 ± 1 °C.

Orange-G equilibrium adsorption isotherms were determined separately at the three different KCl concentrations (0.1, 0.5 and 1.0 M). Isotherm points were obtained by contacting a fixed volume of the feed solutions with the membrane. The fixed volume of the feed solutions was continuously pumped to the membrane cartridge by alternately switching back and forth in a push-pull mode, up to reaching the equilibrium. In contrast, dynamic experiments were performed by continuously pumping the feed solution in one direction until to attain the breakthrough curves.

4. Results and discussion

4.1. Equilibrium adsorption isotherms

The obtained membrane/Orange-G equilibrium adsorption isotherms, in three different KCl solutions, are shown in Fig. 1. The equilibrium points fit well to the Langmuir isotherm (Eq. (7)). As expected, the adsorbed amount of Orange-G decreases with increasing KCl concentration in the liquid-phase. For instant, for a 10^{−3} M concentration of Orange-G in the liquid-phase, the concentration of adsorbed Orange-G in the membrane, was 0.0773, 0.0311 and 0.0166 M when the concentration of KCl in the liquid-phase was 0.1, 0.5 and 1.0 M, respectively. The parameters, C_s^* and b , of the Langmuir isotherm have been obtained by fitting experimental data to the linearized isotherm. These calculated parameters are summarized in Table 1. It is observed that C_s^* values slightly decreased with KCl concentrations, while b values greatly decreased with KCl concentrations. The good fitting of the experimental points to the Langmuir isotherm makes reasonable to assume monolayer adsorption.

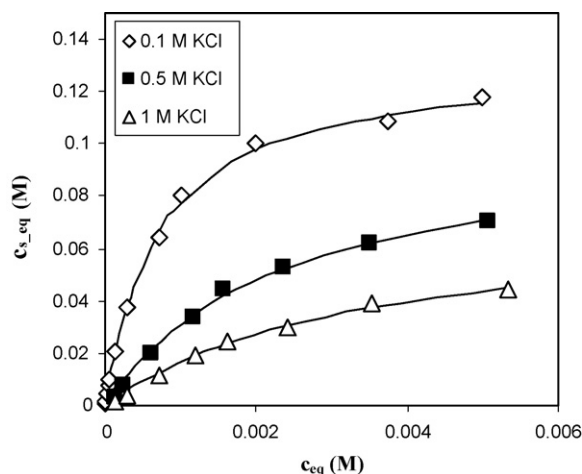


Fig. 1. Adsorption isotherms of Orange-G dye in a Sartobind D ion-exchange membrane adsorber as a function of KCl concentrations.

4.2. Dynamic adsorption

The breakthrough curves of Orange-G depended on two dimensionless parameters (Pe and Ka), which are influenced by the axial dispersion coefficient and the adsorption constant, respectively. In a porous media, the axial molecular diffusion used to have influence on dispersion process only when the molecular Peclet number, Pe_m , is lower than 1:

$$Pe_m = \frac{v \cdot L}{D_m} \quad (29)$$

The diffusivity of an organic compound in water can be estimated by the Hayduk and Laudie method [31]:

$$D_m = \frac{1.326 \times 10^{-8}}{\mu^{1.14} \cdot V^{0.589}} \quad (30)$$

where D_m is the molecular diffusion of the compound in water in $\text{m}^2 \text{s}^{-1}$, μ is the viscosity of water in mPa s (corrected for temperature) and V is the LaBas molar volume in $\text{cm}^3 \text{mol}^{-1}$. V for Orange-G has been estimated to be $600 \text{ cm}^3 \text{mol}^{-1}$ (Cs Chem3D Ultra®, Molecular Modelling and Analysis). Therefore, the molecular diffusion coefficient for Orange-G at 25°C must be about $4.2 \times 10^{-10} \text{ m}^2 \text{s}^{-1}$ and the molecular Peclet number about 83. Therefore, in this case the axial dispersion coefficient characterizes the non-idealities of the membrane adsorber.

The determination of axial dispersion coefficient was conducted experimentally under non-adsorbing conditions, when the flow rate was 1 mL min^{-1} . A cationic membrane adsorber with sulfonic groups was used (Sartobind S). Additionally, dispersion or liquid mixing in the tubing and fittings was also determined by operating in the same conditions but in absence of the membrane adsorber. Fig. 2 shows the breakthrough curves of Orange-G obtained in absence of the membrane adsorber and with the sulfonic membrane. The breakthrough curve corresponding to the experimental device without the membrane adsorber shows an abrupt increase in the outlet concentration with dimensionless time. This means a negligible dispersion produced by the tubing and fittings. In contrast, the breakthrough curve obtained in presence of the mem-

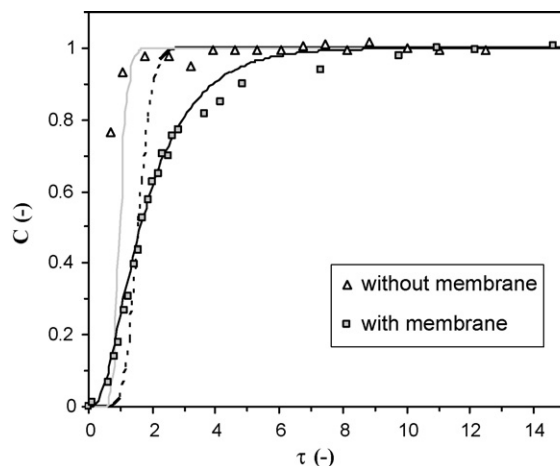


Fig. 2. Experimental breakthrough curves with (squares) and without (triangles) membrane adsorber when the flow rate was 1 mL min^{-1} . Experiments were conducted under non-adsorbing conditions. Solid line and discontinuous line correspond to simulations with $Pe = 2.1$ and $Pe = Pe_m$, respectively. Grey solid line corresponds to simulations without the membrane adsorber. The simulation with $Pe = 2.1$ was conducted assuming small adsorption ($m = 1$ and $Ka = \infty$).

brane becomes gentler due to the dispersion generated by the membrane adsorber. It is expected to detect an outlet dimensionless dye concentration of 0.5 at dimensionless time of 1 in an experiment with a 100% of non-adsorbing conditions. In the experimental breakthrough curve, this concentration was reached at dimensionless time of 1.6, probably due to a relatively small residual adsorption.

Simulations of the experimental breakthrough curve were conducted assuming the presence of a small adsorption. Molecular Peclet number and Peclet number have been tested in the simulations and compared with experimental data (see Fig. 2). The best correlation between simulated and experimental breakthrough was observed with Peclet number equal to 2.1. In order to determine the real axial dispersion coefficient in the membrane, we focused on the dimensionless time elapsed between the 10 and 90% of the dimensionless concentration in the liquid-phase of the breakthrough curve. This time was 3.9 in the experimental breakthrough curve. In the simulations, this time was 3.4 and 0.8 for $Pe = 2.1$ and $Pe = Pe_m = 83$, respectively. Therefore, the real axial dispersion of Orange-G in Sartobind membrane adsorbers is much larger than the molecular dispersion. The reason can be the anisotropic porous structure of the membrane [32].

Many experimental and calculated breakthrough curves were obtained by varying the feed concentrations of Orange-G and KCl and the flow rates. Fig. 3 shows the obtained breakthrough curves for different feed concentrations of Orange-G and KCl, when the flow rate was 1 mL min^{-1} . Fig. 4 shows the obtained breakthrough curves for different concentrations of KCl and flow rates, when the feed concentration of Orange-G was $2.5 \times 10^{-4} \text{ M}$. Table 2 summarizes the fitted parameters of the model for all the experiments. The fitted Peclet number was very similar for all the experiments and was fixed to 2.1 ± 0.1 . The breakthrough curves delay when KCl and dye concentration decrease.

Three key parameters, which characterize the dynamic adsorption process, can be determined from the dimensionless parameters: the axial dispersion coefficient, D_L , from Eq. (12), the adsorption rate constant, k_a , from Eq. (13), and the desorption rate constant, k_d , from Eq. (14). The deduced mean axial dispersion coefficient, D_L , is linearly dependent on flow velocity (Fig. 5), being $(1.29 \pm 0.06) \times 10^{-8} \text{ m}^2 \text{s}^{-1}$ for a flow rate of 1 mL min^{-1} . Fig. 6 shows the adsorption and desorption rate constant as a function of KCl concentration independently of flow rate and feed concentration

Table 1
Langmuir isotherm constants as a function of KCl concentration.

KCl feed concentration (M)	c_s^* (M)	b (M^{-1})
0.1	0.132	1400
0.5	0.104	429
1.0	0.0759	280

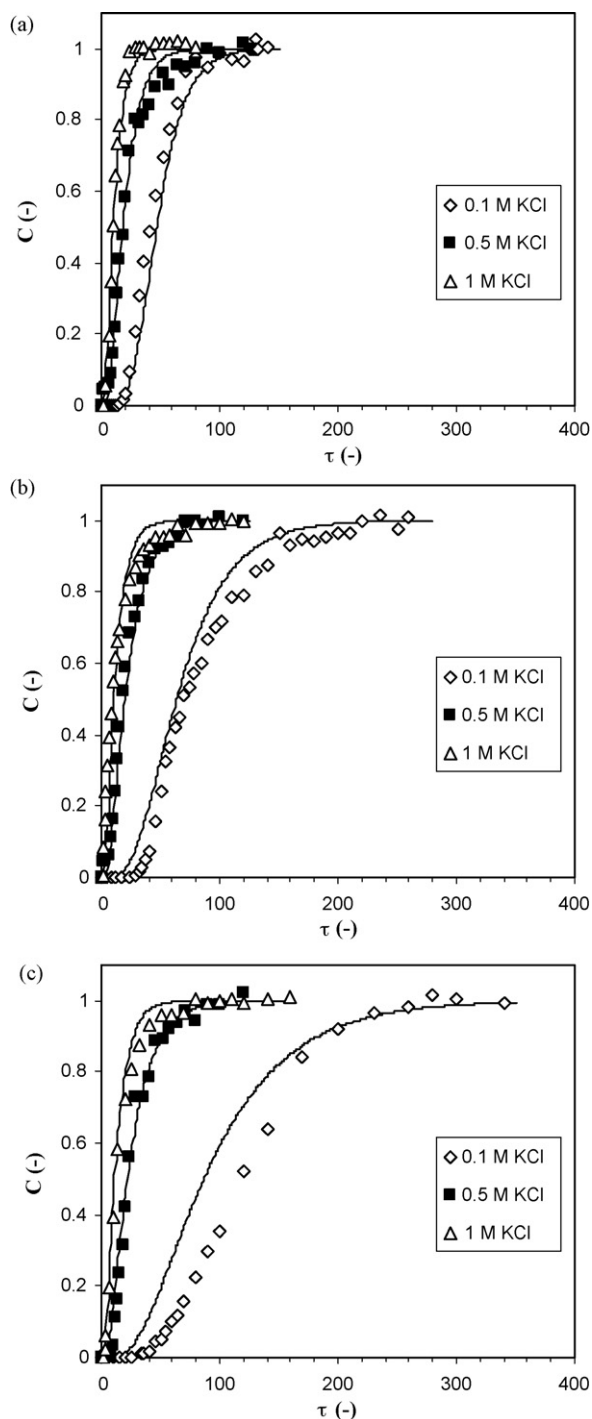


Fig. 3. Experimental and calculated breakthrough curves for different feed concentrations of Orange-G and KCl, when the flow rate was 1 mL min^{-1} . The feed concentrations of Orange-G are (a) 10^{-3} M , (b) $5 \times 10^{-4} \text{ M}$, and (c) $2.5 \times 10^{-4} \text{ M}$. Continuous lines correspond to the theoretical data calculated by fitting the dimensionless dynamic parameter Pe and K_{ads} .

of Orange-G. The adsorption process was hindered in a 61% and desorption process increased in a 33% with increasing the KCl concentration.

The relative importance of the dispersion, adsorption and desorption in the global dynamic process can be assessed by comparing their respective characteristic times with the characteristic time of the process, $t_{process}$. In all the experiments the dispersion time value is twice over the characteristic time of the process, for example, for a flow rate of 1 mL min^{-1} , $t_{process} = 23.6 \text{ s}$ and $t_{disp} = 49.5 \text{ s}$. Then,

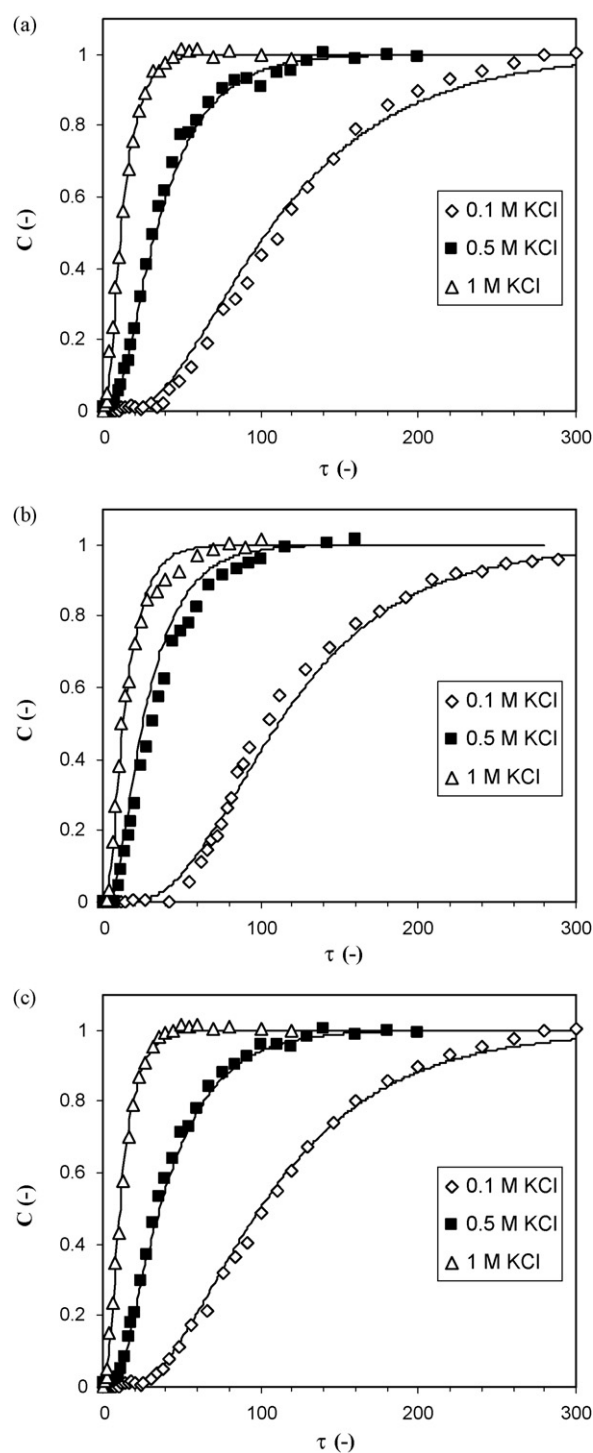


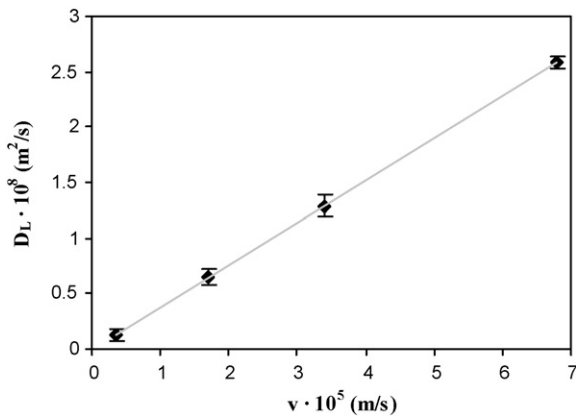
Fig. 4. Experimental and calculated breakthrough curves for different KCl concentration when the feed concentration of Orange-G was $2.5 \times 10^{-4} \text{ M}$ and the flow rates were (a) 2 mL min^{-1} , (b) 0.5 mL min^{-1} , and (c) 0.1 mL min^{-1} .

the dispersion process is significant, since $Pe < 25$, and it cannot be neglected [28]. On the other hand, t_{des} depends on KCl concentration and goes from about 3 to 8 s when KCl concentration decreases from 1.0 to 0.1 M. The desorption process is always important during the dynamic experiments. In contrast, t_{ads} depends on KCl and dye concentration (increasing with KCl concentration and decreasing with dye concentration). t_{ads} goes from 6 s (10^{-3} M dye and 0.1 M KCl) to 38 s ($2.5 \times 10^{-4} \text{ M}$ dye and 1.0 M KCl). Adsorption is not very important at the lower dye concentration ($2.5 \times 10^{-4} \text{ M}$), even when

Table 2

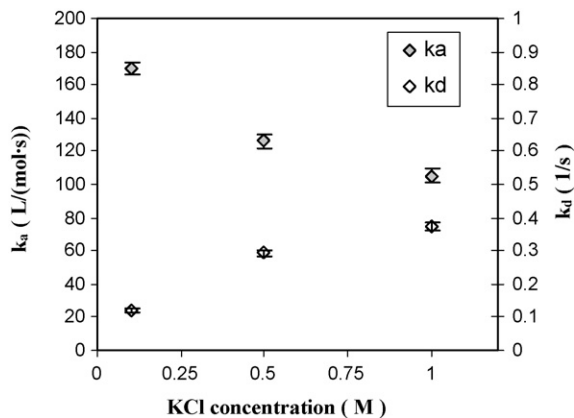
Dimensionless parameters obtained in the simulation of the experimental data. The parameters r , p and m were obtained from the equilibrium data and K_{ads} and $Pe=2.1$ from the dynamic data as a function of flow rate and Orange-G and KCl concentrations.

q (mL min ⁻¹)	c_o (M)	KCl conc. (M)	r	p	m	K_{ads}
1	10 ⁻³	0.1	1.40	0.584	49	4
1	10 ⁻³	0.5	0.429	0.300	20	3
1	10 ⁻³	1.0	0.280	0.219	11	2.5
1	5 × 10 ⁻⁴	0.1	0.701	0.412	70	2
1	5 × 10 ⁻⁴	0.5	0.215	0.177	23	1.4
1	5 × 10 ⁻⁴	1.0	0.140	0.123	12	1.2
1	2.5 × 10 ⁻⁴	0.1	0.351	0.260	88	1
1	2.5 × 10 ⁻⁴	0.5	0.107	0.0969	26	0.71
1	2.5 × 10 ⁻⁴	1.0	0.0700	0.0654	13	0.60
2	2.5 × 10 ⁻⁴	0.1	0.351	0.260	90	0.5
2	2.5 × 10 ⁻⁴	0.5	0.107	0.0969	30	0.37
2	2.5 × 10 ⁻⁴	1.0	0.0700	0.0654	13	0.3
0.5	2.5 × 10 ⁻⁴	0.1	0.351	0.260	110	2
0.5	2.5 × 10 ⁻⁴	0.5	0.107	0.0969	35	1.5
0.5	2.5 × 10 ⁻⁴	1.0	0.0700	0.0654	15	1.2
0.1	2.5 × 10 ⁻⁴	0.1	0.351	0.260	115	10
0.1	2.5 × 10 ⁻⁴	0.5	0.107	0.0969	43	7.5
0.1	2.5 × 10 ⁻⁴	1.0	0.0700	0.0654	15	6

**Fig. 5.** Calculated axial dispersion coefficients as a function of flow velocities.

the KCl is 0.1 M ($t_{ads}=23$ s). Then, the presence of high KCl concentrations hinder the dye adsorption and increase the desorption processes. The anionic dye molecules adsorbed onto diethylamine groups are replaced by chlorine anions.

In general, the modeled breakthrough curves fit well to the experimental data. Only sporadic small deviations can be observed

**Fig. 6.** Calculated adsorption (diamonds) and desorption (squares) rate constants as a function of KCl concentration independently of flow rate and feed concentration of Orange-G.

that can be due to experimental error and membrane irregularities, such as the presence of local dead volumes.

It is important to notice that the parameters obtained for individual solutes, following the previous procedure, can be used to estimate breakthrough curves of blends of these same solutes, without additional experimental work. In this case, we have to solve simultaneously Eq. (15) for each solute and use an extended Langmuir isotherm (a generalization of Eq. (18)):

$$C_{s,eq_i} = \frac{(r_i/p_i) \cdot C_{eq_i}}{1 + \sum_i r_i \cdot C_{eq_i}} \quad (31)$$

Further work will be done to expand this study to include blends.

5. Conclusions

A mathematical model for the dynamic adsorption process through membrane adsorber has been presented in terms of characteristics times for the convection, dispersion, adsorption and desorption elements of the global processes. The relative importance, in the global process, of each element (dispersion, adsorption and desorption) can be assessed by comparing their characteristic times with the convective process time $t_{process}$. It has been observed that the axial dispersion cannot be neglected in the experimental breakthrough curves.

The characteristic dimensionless parameters of the model that controls the equilibrium and dynamic experiments have been separately adjusted. The modeled breakthrough curves of the anionic dye (Orange-G) in the ion-exchange membrane fit well to the experimental. The effect of KCl on the adsorption/desorption process has been measured.

Acknowledgement

The authors are grateful to the Spanish Ministerio de Ciencia y Tecnología (project CTQ2005-08346-C02-01/PPQ) for funds received to carry out this study.

Nomenclature

b	constant of Langmuir equation (M ⁻¹)
c	solute concentration in the liquid-phase (M)
C_{eq}	equilibrium solute concentration in the liquid-phase (M)
C_s	adsorbed solute concentration in the solid-phase (M)
C_s^*	maximum possible concentration of adsorbed solute in the solid-phase (M)
$C_{s,eq}$	equilibrium solute concentration in the solid-phase (M)
$C_{s,eq,0}$	equilibrium feed solute concentration in the solid-phase (M)
C	dimensionless solute concentration in the liquid-phase
C_{eq}	dimensionless equilibrium solute concentration in the liquid-phase
C_s	dimensionless adsorbed solute concentration in the solid-phase
$C_{s,eq}$	dimensionless equilibrium solute concentration in the solid-phase
D_m	axial molecular diffusion (m ² s ⁻¹)
D_L	axial dispersion coefficient (m ² s ⁻¹)
k_a	adsorption rate constant (L mol ⁻¹ s ⁻¹)

k_d	desorption rate constant (s^{-1})
K_{ads}	dimensionless adsorption constant
L	membrane thickness (m)
m	dimensionless ratio between the equilibrium solute concentration in the solid-phase and the feed solute concentration
p	dimensionless ratio between the equilibrium feed solute concentration in the solid-phase and the maximum possible concentration of adsorbed solute in the solid-phase
Pe	axial Peclet number
Pe_m	molecular Peclet number
r	dimensionless ratio between adsorption and desorption constants
t	time (s)
t_{ads}	characteristic time related to adsorption process (s)
t_{des}	characteristic time related to desorption process (s)
t_{disp}	characteristic time related to axial dispersion (s)
$t_{process}$	convective time or characteristic time of the process (s)
V	LaBas molar volume ($cm^3 mol^{-1}$)
v	flow velocity ($m s^{-1}$)
z	axial distance along the membrane (m)

Greek letters

ε	porosity of membrane
μ	viscosity of water (mPa s)
τ	dimensionless time
ζ	dimensionless spatial parameter

References

- [1] G. McKay, J.F. Porter, G.R. Prasad, The removal of dye colours from aqueous solutions by adsorption on low-cost materials, *Water Air Soil Pollut.* 114 (1999) 423.
- [2] H. Yoshida, T. Takemori, Adsorption of direct dye on cross-linked chitosan fiber: breakthrough curve, *Water Sci. Technol.* 35 (1997) 29.
- [3] D.O. Cooney, Adsorption of organic solutes on membrane filters during aqueous phase filtration. 1. Basic rate and equilibrium studies using toluidine blue, *Sep. Sci. Technol.* 27 (1992) 2001.
- [4] C. Namasivayam, R. Jeyakumar, R.T. Yamura, Dye removal from waste-water by adsorption waste Fe(III)/Cr(III) hydroxide, *Waste Manage.* 14 (1994) 643.
- [5] R.K. Lewus, G. Carta, Binary protein adsorption on gel-composite ion-exchange media, *AIChE J.* 45 (1999) 512.
- [6] B.J. Horstmann, H.A. Chase, Modeling the affinity adsorption of immunoglobulin-G to protein-A immobilized to agarose matrices, *Chem. Eng. Res. Des.* 67 (1989) 243.
- [7] G.L. Skidmore, B.J. Horstmann, H.A. Chase, Modeling single-component protein adsorption to the cation exchanger Sepahrose FF, *J. Chromatogr.* 498 (1990) 113.
- [8] C. Boi, F. Cattoli, R. Facchini, Adsorption of lectins on affinity membranes, *J. Membr. Sci.* 273 (2006) 12.
- [9] W. Guo, E. Ruckenstein, Separation and purification of horseradish peroxidase by membrane affinity chromatography, *J. Membr. Sci.* 211 (2003) 101.
- [10] D.K. Roper, E.N. Lightfoot, Separation of biomolecules using adsorptive membrane, *J. Chromatogr. A* 702 (1995) 3.
- [11] R. Ghosh, Protein separation using membrane chromatography: opportunities and challenges, *J. Chromatogr. A* 952 (2002) 13.
- [12] Q.D. Lan, A.S. Bassi, J.X. Zhu, A. Margaritis, A modified Langmuir model for the prediction of the effects of ionic strength on the equilibrium characteristics of protein adsorption onto ion exchange/affinity adsorbents, *Chem. Eng. J.* 81 (2001) 179.
- [13] Z. Ma, K. Masaya, S. Ramakrishna, Immobilization of Cibacron blue F3GA on electrospun polysulphone ultra-fine fiber surfaces towards developing an affinity membrane for albumin adsorption, *J. Membr. Sci.* 282 (2006) 237.
- [14] M.Y. Arica, H.N. Testereci, A. Denizli, Dye-ligand and metal chelate poly(2-hydroxyethylmethacrylate) membranes for affinity separation of proteins, *J. Chromatogr. A* 799 (1998) 83.
- [15] S.P. Zhang, Y. Sun, Steric mass-action model for dye-ligand affinity adsorption of protein, *J. Chromatogr. A* 957 (2002) 89.
- [16] S.P. Zhang, Y. Sun, Study on protein adsorption kinetics to a dye-ligand adsorbent by the pore diffusion model, *J. Chromatogr. A* 964 (2002) 35.
- [17] T. Vicente, M.F.Q. Sousa, C. Peixoto, J.P.B. Mota, P.M. Alves, M.J.T. Carrondo, Anion-exchange chromatography for purification of rotavirus-like particles, *J. Membr. Sci.* 311 (2008) 270.
- [18] M.Y. Arica, M. Yilmaz, E. Yalcin, G. Bayramoglu, Affinity membrane chromatography: relationship of dye-ligand type to surface polarity and their effect on lysozyme separation and purification, *J. Chromatogr. B* 805 (2004) 315.
- [19] H.L. Knudsen, R.L. Fahrner, Y. Xu, L.A. Norling, G.S. Blank, Membrane ion-exchange chromatography for process-scale antibody purification, *J. Chromatogr. A* 907 (2001) 145.
- [20] A. Tejeda-Mansir, R.M. Montesinos, R. Guzman, Mathematical analysis of frontal affinity chromatography in particle and membrane configurations, *J. Biochem. Biophys. Methods* 49 (2001) 1.
- [21] R.M. Montesinos, A. Tejeda-Mansir, R. Guzman, J. Ortega, W.E. Schiesser, Analysis and simulation of frontal affinity chromatography of proteins, *Sep. Purif. Technol.* 42 (2005) 75.
- [22] A.I. Liapis, K.K. Unger, G. Steet, Highly Selective Separations in Biotechnology, Chapman and Hall, Glasgow, NZ, 1994.
- [23] S.Y. Suen, M.R. Etzel, A mathematical analysis of affinity membrane bioseparations, *Chem. Eng. Sci.* 47 (1992) 1355.
- [24] M. Sahimi, A.O. Imdakm, The effect of morphological disorder on hydrodynamic dispersion in flow through porous media, *J. Phys. A: Math. Gen.* 21 (1988) 3833.
- [25] W. Shi, F. Zhang, G. Zhang, Mathematical analysis of affinity membrane chromatography, *J. Chromatogr. A* 1081 (2005) 156.
- [26] C. Boi, S. Dimartino, G.C. Sarti, Modelling and simulation of affinity membrane adsorption, *J. Chromatogr. A* 1162 (2007) 24.
- [27] C. Boi, Membrane adsorbents as purification tools for monoclonal antibody purification, *J. Chromatogr. B* 848 (2007) 19.
- [28] H.C. Liu, J.R. Fried, Breakthrough of lysozyme through an affinity membrane of cellulose-cibacron blue, *AIChE J.* 40 (1994) 40.
- [29] C. Tien, Adsorption Calculations and Modelling, Butterworth-Heinemann, Boston, 1994, p. 140.
- [30] I. Tatárova, R. Fáber, R. Denoyel, M. Polakovic, Characterization of pore structure of a strong anion-exchange membrane adsorbent under different buffer and salt concentration conditions, *J. Chromatogr. A* 1216 (2009) 941.
- [31] W. Hayduk, H. Laudie, Prediction of diffusion coefficients for nonelectrolytes in dilute aqueous solutions, *AIChE J.* 20 (1974) 611.
- [32] J. Wang, F. Dismer, J. Hubbuch, M. Ulbricht, Detailed analysis of membrane adsorber pore structure and protein binding by advanced microscopy, *J. Membr. Sci.* 320 (2008) 456.